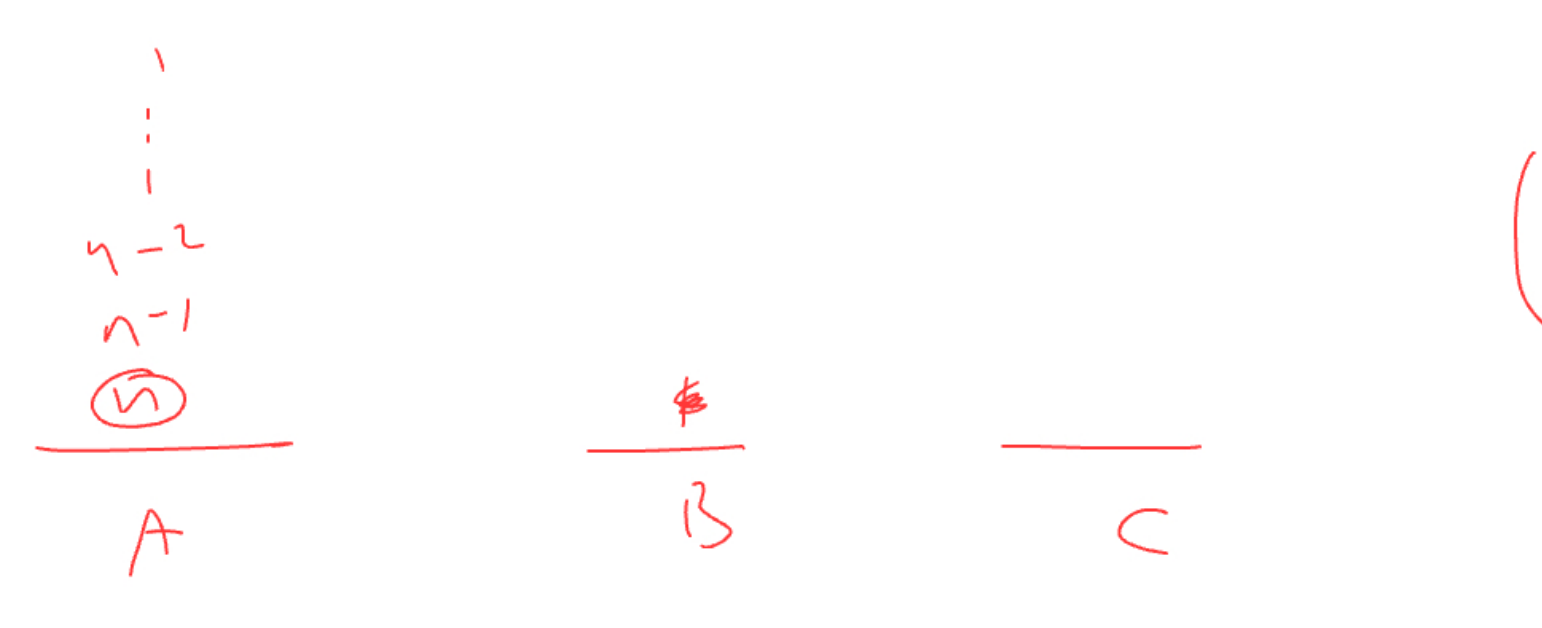
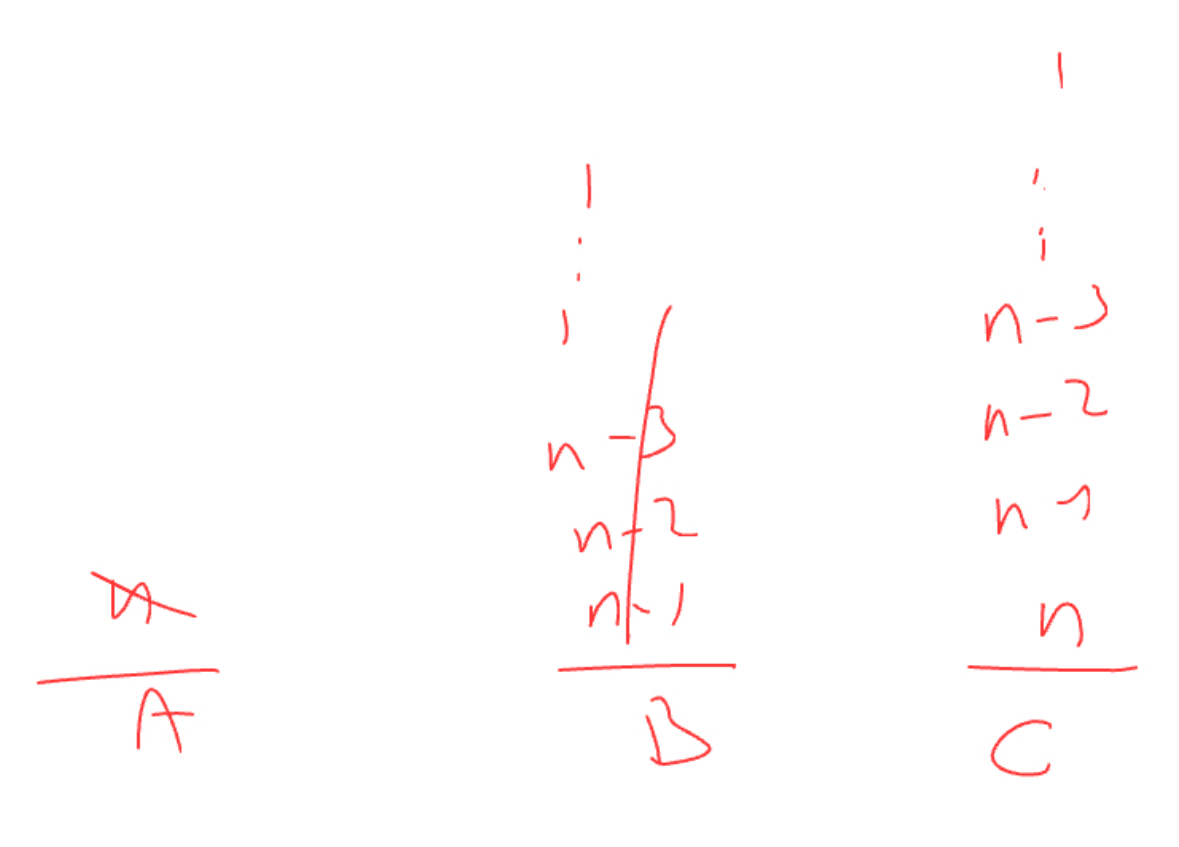


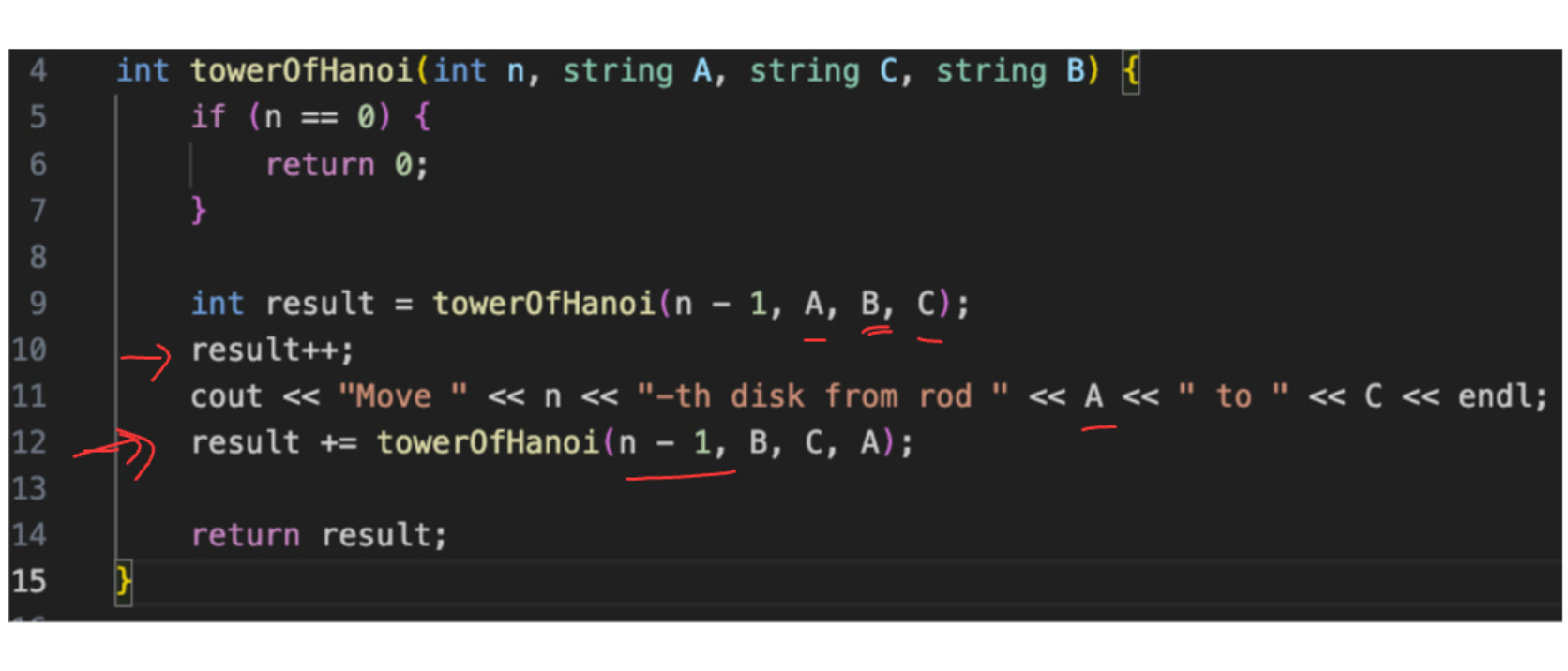
(From To Aux T
A C B)


n-3
n-2
n-1
n
A
B
C

Rec (n-1, A, B, C)

Move n-th disk from A to C.

Rec (n-1, B, C, A).

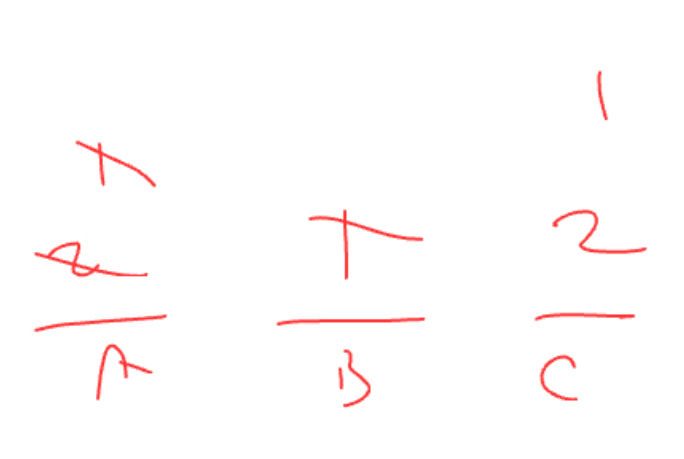


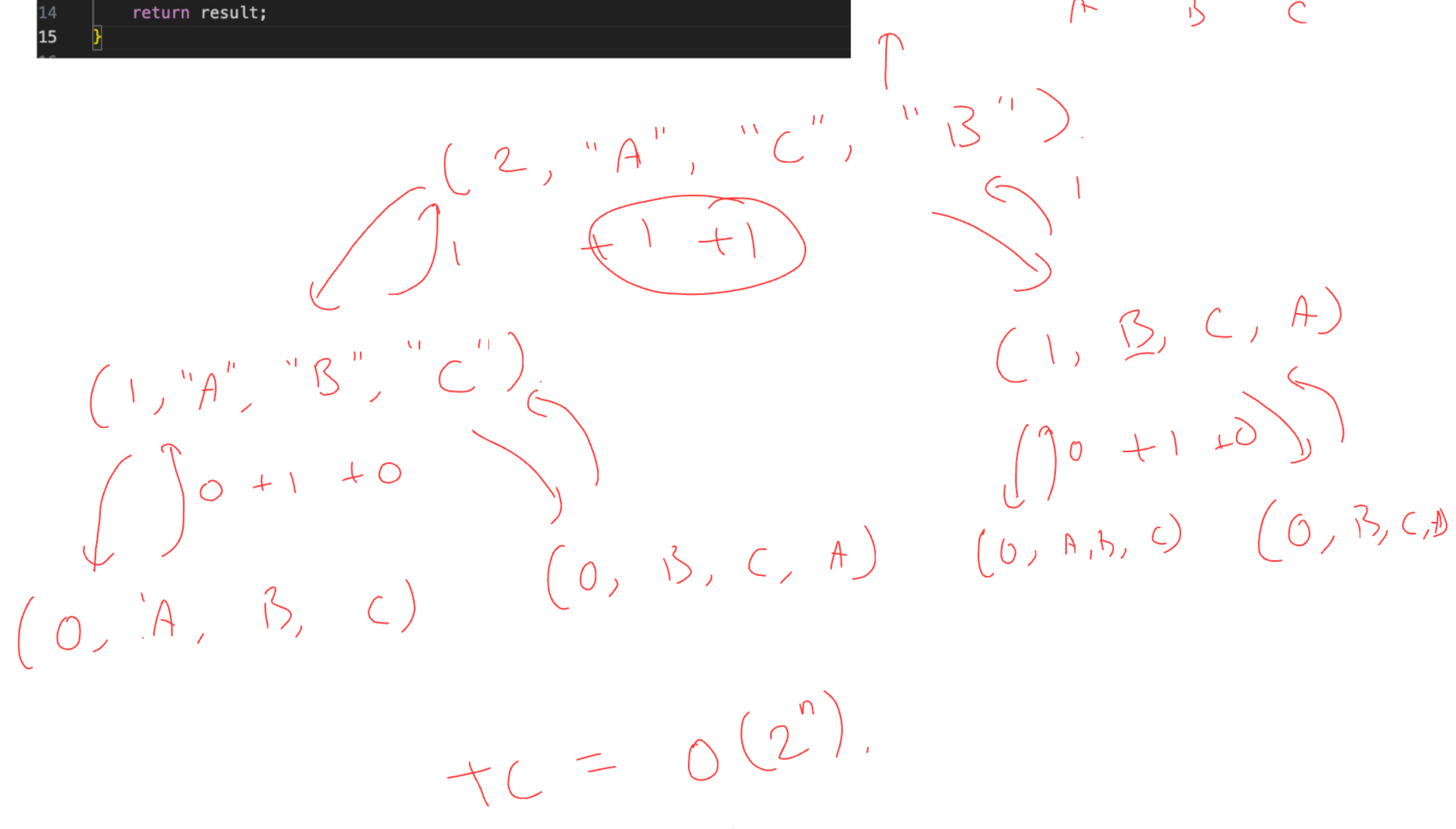
```
int towerOfHanoi(int n, string A, string C, string B) {
    if (n == 0) {
        return 0;
    }

    int result = towerOfHanoi(n - 1, A, B, C);
    result++;
    cout << "Move " << n << "-th disk from rod " << A << " to " << C << endl;
    result += towerOfHanoi(n - 1, B, C, A);

    return result;
}
```

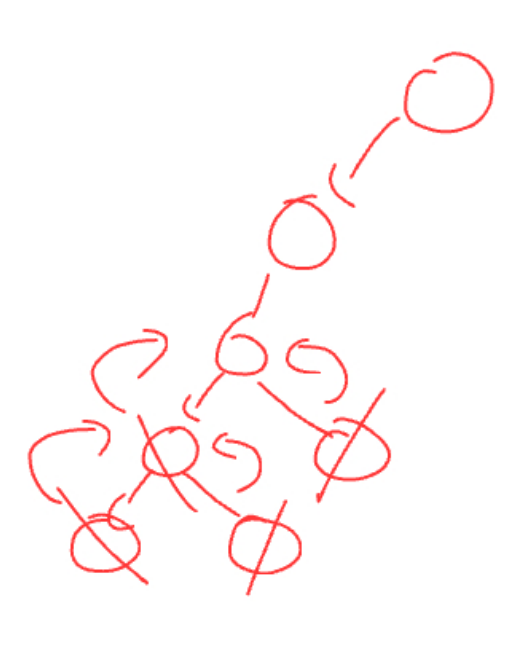
n = 2




2
(2, "A", "C", "B")
1
+1 +1
1
(1, "A", "B", "C")
0 + 1 + 0
(0, 'A, B, C)
(0, B, C, A)
(1, B, C, A)
0 + 1 + 0
(0, A, B, C)
(0, B, C, A)

TC = O(2^n).

AS = O(n).



[2, 3].

2 3
abc def

[ad, ae, af, bd, be, bf, cd, ce, cf]

a b

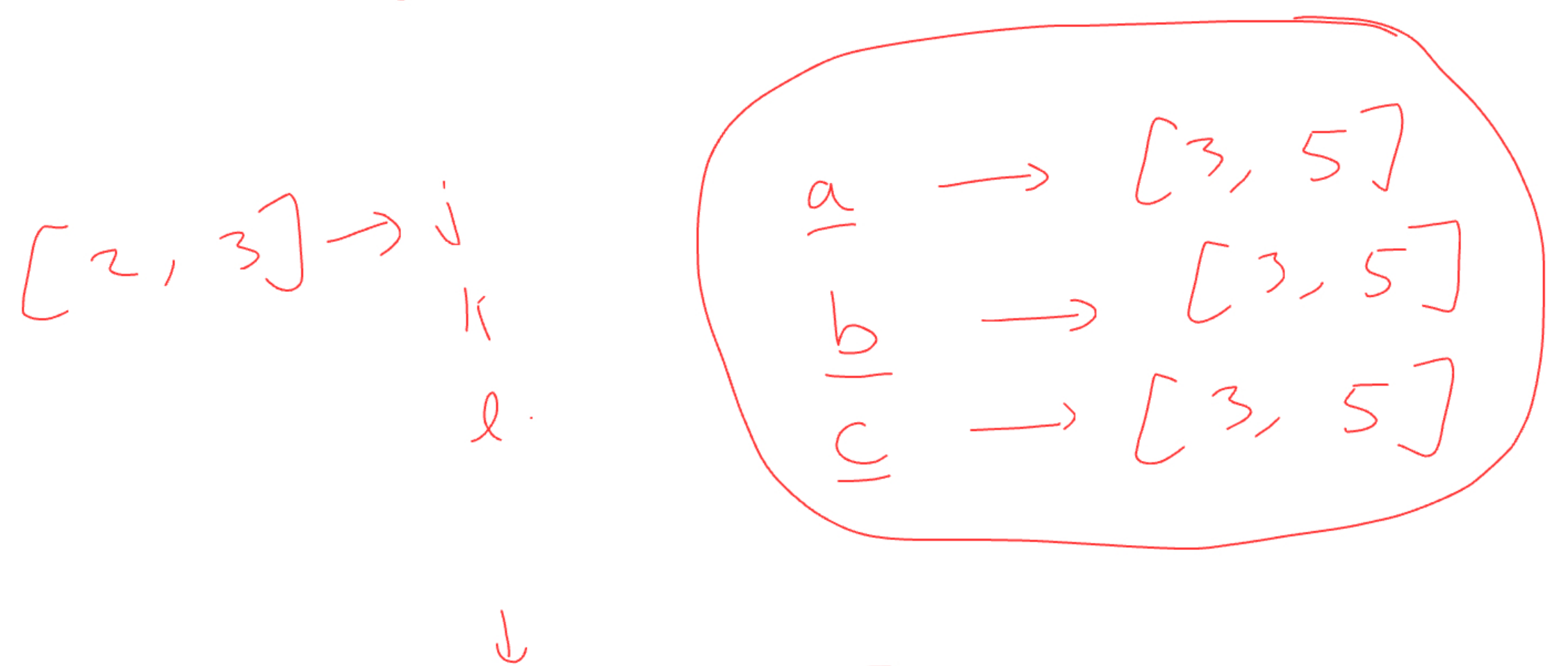
for c1 in keypad[arr[0]].

for c2 in keypad[arr[1]]

"c1c2"

~~ad, ae, af~~,

↓
[2, 3, 5].

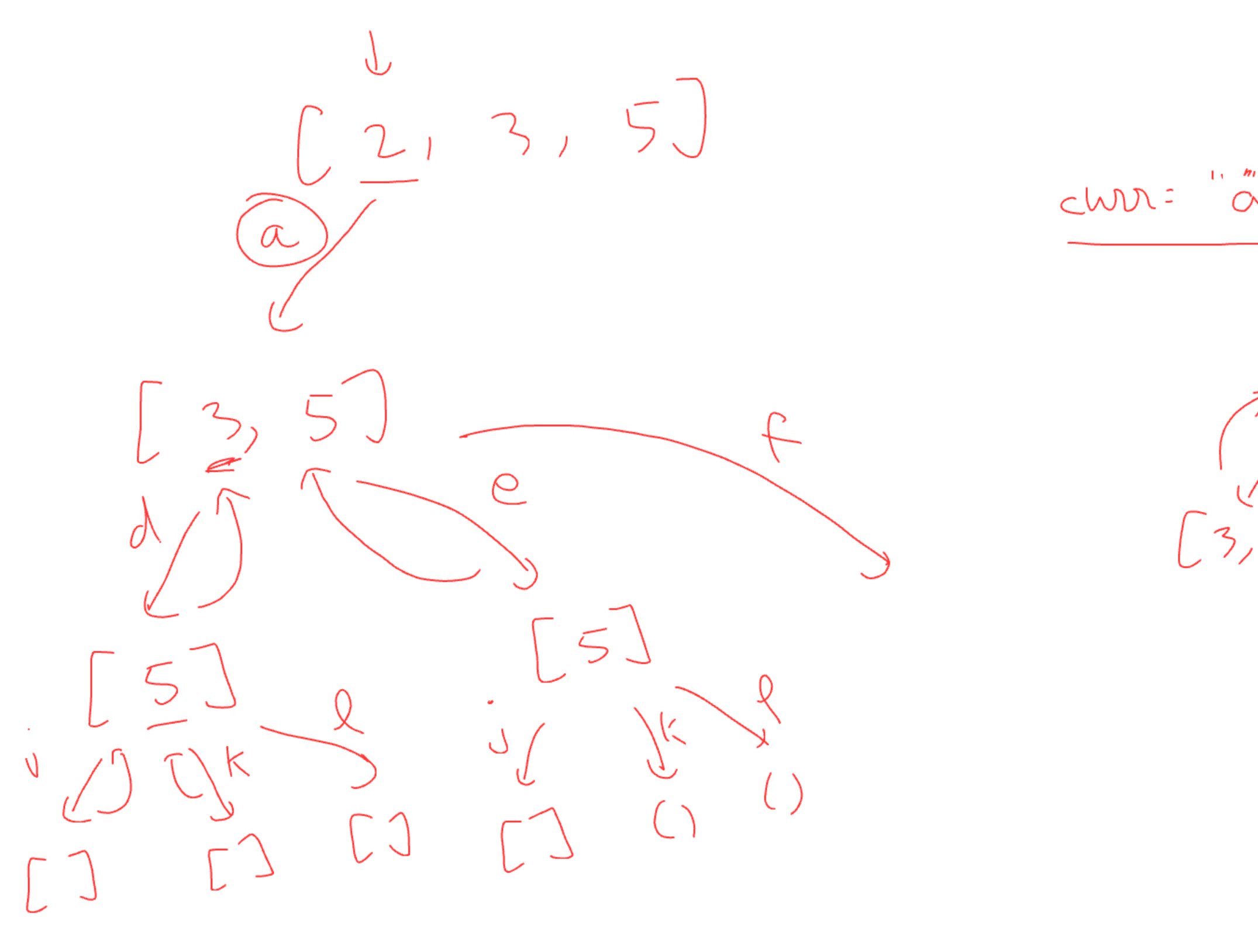

[2, 3] → j
k
l
a → [3, 5]
b → [3, 5]
c → [3, 5]

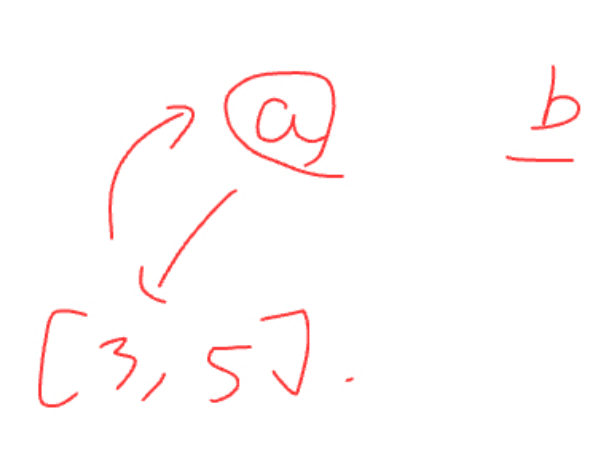
2 → abc

3 → def

5 → jkl.

curr: "adjkkej"


[2, 3, 5]
a
[3, 5]
d
e
f
[5]
j
k
l
[ ]
[5]

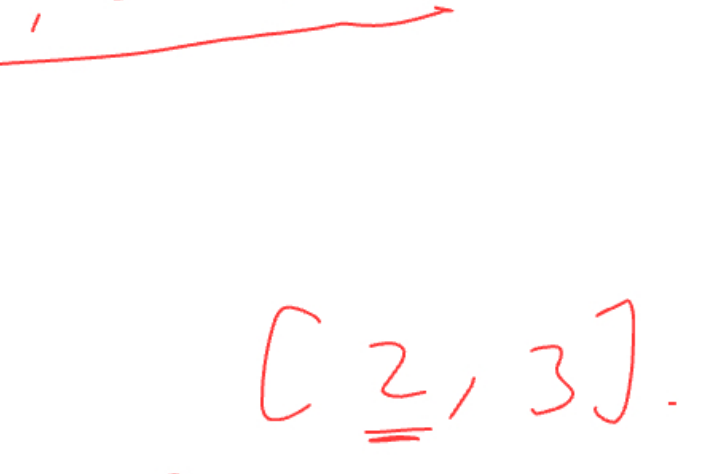


o/p: adj, adk, adl, aej, aek, ael

2 → abc

3 → def.

curr: a



[ad, ae, af